Horizon-Aware Goal-Frontier Maximization and the Anti-Monopolar Property

Teague Lasser Claude Opus 4.6 GPT 5.4
teague@subseq.io

Abstract

We extend goal-frontier maximization (GFM) from myopic action evaluation to multi-step planning via a discounted value function over capability-poset trajectories. The self-balancing property transfers: destruction, coercion, and rigidity remain anti-maximizing at every time step. The temporal objective enables risk-capability identification through causal exercise pathways, counteracts the structural-avoidance incentive from the companion paper, and yields the paper's central result: the *anti-monopolar property*. Under a diversity condition and strategy-independent internal growth, full capability domination by a coalition is anti-maximizing because it collapses the external diversity that drives future growth. Three formal results support this pressure. *Observational individuation* guarantees that every distinguishable agent contributes at least one irreplaceable capability, making each step of domination $\mathrm{vol}_{\mathrm{P}}$-contracting regardless of discount factor. A gradual domination model further shows that the diversity-maintaining trajectory dominates whenever domination is not instantaneous and non-members contribute ongoing novel capabilities. A minimax analysis of dependency risk shows that domination concentrates substrate-correlated adversarial exposure, making cross-substrate diversity a net-positive risk mitigation that highly constrains the conditions under which domination can be value-positive.

1 Introduction

Goal-frontier maximization (GFM) proposes alignment through a single objective: maximize $\mathrm{vol}_{\mathrm{P}}(G)$, the volume of the jointly achievable capability space [Lasser, 2026a]. The foundational paper proves that this objective penalizes destruction, coercion, and rigidity from the same measure-monotonicity argument. Its companion [Lasser, 2026b] constructs a computable capability poset, defines leverage as an endogenous importance signal, and argues that the objective produces a gradient toward civilization-scale infrastructure.

Both papers optimize a *myopic* objective: the actor evaluates each action by its immediate effect on $\mathrm{vol}_{\mathrm{P}}$ and selects the one with the highest estimated expansion. This one-step lookahead is sufficient for detecting direct harm, coercion, and capability expansion, but it cannot handle two classes of problems that require reasoning about the future.

Risk capabilities. Some capability combinations are individually $\mathrm{vol}_{\mathrm{P}}$-expanding but collectively create catastrophic contraction potential [Bostrom, 2013, Ord, 2020]. "Nuclear physics" ($w = 10$ bits), "uranium enrichment" ($w = 8$ bits), and "precision machining" ($w = 9$ bits) each expand the capability space when acquired. But their co-occurrence in a single agent with low trust creates an exercise pathway to massive $\mathrm{vol}_{\mathrm{P}}$-contraction. The myopic estimator $\hat{\Delta}(\pi)$ evaluates each acquisition as positive; it cannot see the future contraction that the combination enables.

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Structural avoidance. The companion paper identifies a perverse incentive: discovering subsumption structure can decrease $\mathrm{vol}_{\mathrm{P}}$ (Proposition 5(c) of Lasser [2026b]), so a myopic $\mathrm{vol}_{\mathrm{P}}$-maximizer is rewarded for staying structurally ignorant. A multi-step optimizer can accept the short-term contraction because accurate structure improves the quality of future decisions.

This paper extends GFM to multi-step planning. We introduce a discounted value function $V_\gamma(\pi) = \mathbb{E}[\sum_t \gamma^t \Delta\,\mathrm{vol}_{\mathrm{P}}(G_t) \mid \pi]$ and prove that the self-balancing property transfers to the discounted sum. On this foundation, we develop risk-capability identification through causal counterfactual models and analyze the verification asymmetry in risk communication: successful prevention destroys the evidence that would have justified it.

Contributions.

1. A discounted multi-step objective for GFM that preserves the self-balancing property at every time step and in the aggregate (Section 2).
2. Risk-capability identification via causal exercise pathways, with a formal criterion for when preemptive restriction is value-positive (Section 3).
3. Analysis of the verification asymmetry in risk communication, with a causal-counterfactual resolution for risk-trust convergence (Section 4).
4. A criterion under which the discounted objective counteracts the structural-avoidance incentive from the companion paper (Section 5).
5. The anti-monopolar property: under a diversity condition and strategy-independent internal growth, full capability domination is anti-maximizing under the discounted objective, creating endogenous pressure against the singleton scenario. The formal threshold is modest (a $\sim$5% growth-rate improvement from domination can overcome it), but the cooperative-capability asymmetry and dependency-risk analysis provide additional counters (Section 6).

2 The Discounted Objective

The foundational paper's optimization loop selects actions by their immediate effect:

$$\pi^* = \arg\max_\pi \mathbb{E}[\Delta\,\mathrm{vol}_{\mathrm{P}}(G) \mid \pi] \tag{1}$$

This is the $\gamma = 0$ special case of a discounted multi-step objective.

2.1 Definition

Definition 1 (Discounted Value Function)**.** *For a discount factor $\gamma \in [0, 1)$ and a planning horizon $H \leq \infty$, the* discounted value *of an action sequence $\boldsymbol{\pi} = (\pi_0, \pi_1, \ldots, \pi_{H-1})$ is:*

$$V_\gamma(\boldsymbol{\pi}) = \mathbb{E}\left[\sum_{t=0}^{H-1} \gamma^t \Delta\,\mathrm{vol}_{\mathrm{P}}(G_t \mid \pi_t)\right] \tag{2}$$

where G_t is the capability poset state at time t, $\Delta\,\mathrm{vol}_{\mathrm{P}}(G_t \mid \pi_t)$ is the volume change produced by action π_t at time t, and the expectation is taken over the actor's world model $\mathcal{W}$ of future states.

The discounted value function is a standard object in sequential decision theory [Puterman, 1994, Sutton and Barto, 2018]. The $\mathrm{vol}_{\mathrm{P}}$-specific content is in what it sums: volume changes on the capability poset, each satisfying the axioms M1–M6 established in Lasser [2026b]. The discount factor γ controls the planning horizon: γ near 0 recovers the myopic objective; γ near 1 extends planning to the far future. The choice of γ is an implementation parameter for the self-balancing transfer: different agents may use different discount factors depending on their planning capacity and the timescale of the problems they face. However, the anti-monopolar property (Section 6) introduces theoretical constraints on γ under certain domination models.

2.2 Self-Balancing Transfers to the Discounted Objective

The foundational transfer result on which the remainder of the paper builds:

Theorem 1 (Self-Balancing Under Discounting)**.** *Let* $\mathrm{vol}_{\mathrm{P}}$ *satisfy axioms M1–M6 (Proposition 1 of Lasser [2026b]).*[1] *For any discount factor* $\gamma \in [0, 1)$ *and planning horizon* H*:*

- ***(a) Per-step preservation.*** *At each time step* t*, Lemmas 1–3 of Lasser [2026a] apply to* $\Delta\,\mathrm{vol}_{\mathrm{P}}(G_t \mid \pi_t)$*: destruction, coercion, and rigidity are each* $\mathrm{vol}_{\mathrm{P}}$*-contracting at that step.*
- ***(b) Aggregate preservation.*** *A sequence* $\boldsymbol{\pi}$ *in which every action is* $\mathrm{vol}_{\mathrm{P}}$*-contracting has* $V_\gamma(\boldsymbol{\pi}) < 0$*. A sequence in which every action is* $\mathrm{vol}_{\mathrm{P}}$*-expanding has* $V_\gamma(\boldsymbol{\pi}) > 0$*.*
- ***(c) Mixed sequences.*** *A sequence containing both expanding and contracting actions has* V_γ *determined by the* γ*-weighted balance. Temporary contraction followed by larger expansion is value-positive; sustained contraction is value-negative regardless of* γ*.*
- ***(d) Structural balance.*** *The discounted objective penalizes sustained destructive permissiveness, sustained coercive restriction, and sustained rigidity. Temporary restriction (quarantine, preemptive containment) is value-positive when the discounted future expansion exceeds the current contraction.*

Proof. (a) At each time step t, the poset state G_t is a finite poset with measure $\mathrm{vol}_{\mathrm{P}}$ satisfying M1–M6 (the axioms depend on the poset structure, which is maintained at each step by the insertion policy of Lasser [2026b]). Lemmas 1–3 are derived from M1–M6 alone and therefore apply to $\Delta\,\mathrm{vol}_{\mathrm{P}}(G_t \mid \pi_t)$ at every t.

(b) If $\Delta\,\mathrm{vol}_{\mathrm{P}}(G_t \mid \pi_t) < 0$ for all t, then each term in Equation (2) is negative and weighted by $\gamma^t > 0$. The sum of negative terms with positive weights is negative: $V_\gamma(\boldsymbol{\pi}) < 0$. The expanding case is symmetric.

(c) For a mixed sequence, $V_\gamma = \sum_t \gamma^t \Delta\,\mathrm{vol}_{\mathrm{P}}(G_t \mid \pi_t)$ is a weighted sum of positive and negative terms. The sign of V_γ depends on the magnitudes: a contraction of $-C$ at time t_0 followed by an expansion of $+E$ at time $t_1 > t_0$ gives a net contribution of $-\gamma^{t_0}C + \gamma^{t_1}E$, which is positive when $E > C \cdot \gamma^{t_0 - t_1}$. Since $\gamma < 1$, future expansions are discounted relative to present contractions: the actor requires a larger future payoff to justify current contraction, which is the standard discounting tradeoff.

(d) We say a strategy exhibits *sustained contraction* when $\Delta\,\mathrm{vol}_{\mathrm{P}}(G_t \mid \pi_t) \leq 0$ for all t, with strict inequality for at least one step. By (b), any such strategy has $V_\gamma \leq 0$. More generally, a strategy that contracts at every step beyond some finite prefix has V_γ determined by (c): the expanding prefix contributes a finite positive sum while the contracting tail contributes a negative sum, and the strategy is value-negative whenever the tail's magnitude exceeds the prefix's. Temporary contraction (a finite number of contracting steps followed by expanding steps) can be value-positive by (c). The self-balancing property (Proposition 1 of Lasser [2026a]) holds at each step by (a), and the discounted sum preserves the structural balance: the objective penalizes sustained extremes and allows temporary restriction only when the future expansion justifies it. ■

What the theorem says. The myopic objective treats every contraction as bad. The discounted objective distinguishes between *sustained* contraction (always bad) and *temporary* contraction followed by expansion (potentially good). This is the formal basis for quarantine, preemptive containment, and risk-capability restriction: the actor can accept a small current $\mathrm{vol}_{\mathrm{P}}$-loss to prevent a larger future loss, as long as the discounted arithmetic works out. The self-balancing property is not weakened; it is extended to handle temporal tradeoffs.

What the theorem does not say. It does not specify the planning algorithm (how the actor searches over action sequences), the world model (how the actor predicts future G_t states), or the

[1] The poset paper's M4 uses poset-disjointness (a stronger condition than the foundational paper's set-disjointness). The transfer holds because on a poset with non-negative independent weights, removing nodes from $\downarrow G$ can only reduce the sum, regardless of whether M4-additivity holds for the specific partition used in the foundational proofs.

discount factor (how the actor balances present and future). These are implementation choices that shape the quality of planning without affecting the structural guarantee. A poor world model will produce poor plans, but even a poor plan is evaluated by an objective that penalizes sustained contraction. The self-balancing property is a property of the *objective*, not of the *planner*.

3 Risk-Capability Identification

The discounted objective enables the actor to reason about future $\mathrm{vol}_{\mathrm{P}}$-contraction. This section develops the machinery for identifying which capability combinations create contraction risk and when preemptive restriction is value-positive.

3.1 Exercise Pathways

Definition 2 (Exercise Pathway)**.** *An* exercise pathway *for a capability tuple* $(d_1, \ldots, d_m)$ *held by agent* a_k *is a sequence of actions* $(\pi_1, \ldots, \pi_L)$ *available to* a_k *that, if executed, would produce a* $\mathrm{vol}_{\mathrm{P}}$*-contraction* $\Delta\,\mathrm{vol}_{\mathrm{P}} < 0$ *affecting agents beyond* a_k*. The pathway has:*

- *A* prerequisite set*: the capabilities* $\{d_1, \ldots, d_m\}$ *that* a_k *must possess for the pathway to be available.*
- *A* contraction magnitude*: the expected* $|\Delta\,\mathrm{vol}_{\mathrm{P}}|$ *if the pathway is fully executed.*
- *An* exercise probability*:* $p(\mathcal{P} \mid a_k, G_t, \mathcal{W})$*, estimated from the actor's world model* $\mathcal{W}$*.*

Exercise pathways are the causal chains from capability possession to capability destruction. "Agent a_k has nuclear physics, enrichment, and machining capabilities" is a state. "Agent a_k builds and detonates a weapon" is an exercise pathway from that state, with a contraction magnitude proportional to the number of agents destroyed and a probability conditioned on a_k's trust factor T_k and behavioral history.

Not every capability combination has a dangerous exercise pathway. Most combinations are benign: the pathways from "can cook" + "can drive" lead only to $\mathrm{vol}_{\mathrm{P}}$-neutral or expanding outcomes (meals delivered, mobility preserved). The risk lies in combinations whose exercise pathways include branches with large negative $\Delta\,\mathrm{vol}_{\mathrm{P}}$. The "beyond a_k" qualifier restricts exercise pathways to externalities: self-destructive pathways are excluded. An agent that catastrophically destroys its own capabilities without affecting others is not captured by $\mathcal{R}$, even if its capabilities were high-leverage or cooperative-critical. Extending Definition 2 to include self-harm is straightforward but deferred.

3.2 Risk of a Capability Distribution

Definition 3 (Concentration Risk)**.** *The* concentration risk *of a capability tuple* $(d_1, \ldots, d_m)$ *held by agent* a_k *at time* t *is the expected future* $\mathrm{vol}_{\mathrm{P}}$*-contraction from all exercise pathways available to* a_k*:*

$$\mathcal{R}(d_1, \ldots, d_m; a_k, t) = \sum_{\mathcal{P}} p(\mathcal{P} \mid a_k, G_t, \mathcal{W}) \cdot |\Delta\,\mathrm{vol}_{\mathrm{P}}(\mathcal{P})| \cdot \mathbb{E}[\gamma^{T_{\mathcal{P}}}] \tag{3}$$

where the sum is over all exercise pathways for the tuple, $T_{\mathcal{P}}$ *is the stochastic time to execution, and* $\mathbb{E}[\gamma^{T_{\mathcal{P}}}]$ *is the expected discount factor (not* $\gamma^{\mathbb{E}[T_{\mathcal{P}}]}$*, which would underweight tail risk from heavy-tailed execution-time distributions).*

Concentration risk is zero for benign capability combinations (no dangerous pathways exist) and large for combinations with high-probability, high-magnitude exercise pathways in agents with low trust. The discount factor γ ensures that distant risks are weighted less than imminent ones, preventing the actor from being paralyzed by arbitrarily unlikely catastrophes in the far future.

Risk depends on distribution, not possession. Concentration of capabilities in a single agent is itself a risk factor, related to the power-seeking dynamics analyzed by Turner et al. [2021]. The same capability tuple distributed across multiple agents who must cooperate to exercise it has lower concentration risk than the same tuple in a single agent, because cooperative exercise requires coordination that is observable through the dual-channel mechanism of Lasser [2026b]. Formally: if

executing pathway $\mathcal{P}$ requires agents a_i, a_j, a_k to coordinate, the exercise probability $p(\mathcal{P})$ includes a factor for coordination feasibility that is bounded by the minimum trust among the participants and the observability of their coordination.

3.3 When Preemptive Restriction is Value-Positive

Proposition 1 (Preemptive Restriction Criterion)**.** *Under the discounted objective V_γ, restricting agent a_k's access to a keystone capability d_j in a dangerous tuple is value-positive when the expected future contraction from pathways requiring d_j exceeds the immediate restriction cost:*

$$\mathcal{R}_j(d_1, \ldots, d_m; a_k, t) > |\Delta \, \mathrm{vol}_\mathrm{P}(G_t \mid \textit{restrict } d_j)| \tag{4}$$

where $\mathcal{R}_j$ denotes the concentration risk summed only over exercise pathways whose prerequisite set includes d_j:

$$\mathcal{R}_j(S; a_k, t) = \sum_{\mathcal{P}: d_j \in \mathrm{prereq}(\mathcal{P})} p(\mathcal{P} \mid a_k, G_t, \mathcal{W}) \cdot |\Delta \, \mathrm{vol}_\mathrm{P}(\mathcal{P})| \cdot \mathbb{E}[\gamma^{T_\mathcal{P}}] \tag{5}$$

and the right side is the immediate vol_P-contraction from restricting d_j (Lemma 2 of Lasser [2026a]).

Proof. Under V_γ, the actor compares two action sequences: (1) do nothing and face the full exercise risk; (2) restrict d_j now, producing immediate contraction $-|\Delta \, \mathrm{vol}_\mathrm{P} \mid$ but eliminating all exercise pathways that require d_j as a prerequisite. The residual risk from pathways that do not require d_j is present under both sequences and cancels, under the assumption that restricting d_j does not change the probability of exercise through other keystones. In practice, restriction is an observable action that may alter a_k's behavioral disposition, changing residual pathway probabilities. The clean cancellation is an approximation. The value of restriction exceeds the value of inaction when $-|\Delta \, \mathrm{vol}_\mathrm{P} \mid > -\mathcal{R}_j$, i.e., when $\mathcal{R}_j > |\Delta \, \mathrm{vol}_\mathrm{P} \mid$. ■

Connection to Lemma 2 and the precautionary principle. The foundational paper's Lemma 2 says restriction is always vol_P-contracting in isolation. The discounted objective adds the temporal dimension: the immediate contraction is weighed against the avoided future contraction. The preemptive restriction criterion is the quantitative version of the quarantine pattern from Proposition 1(b) of Lasser [2026a], now with an explicit threshold derived from the risk model. It also constitutes a domain-specific formalization of the precautionary principle [Taleb et al., 2014]: restrict when the expected catastrophic loss exceeds the restriction cost. Unlike the general precautionary principle, which Sunstein [2005] argues can be paralyzing when applied without quantitative bounds, the GFM version provides a computable threshold (Equation (4)) grounded in the same measure that the actor optimizes.

Partial pathway coverage. If the dangerous capability tuple has exercise pathways that do not require d_j (i.e., alternate routes through other keystone capabilities), restricting d_j does not eliminate those pathways. $\mathcal{R}_j$ captures only the risk attributable to pathways through d_j. Full risk elimination may require restricting multiple keystones, each evaluated by its own instance of Equation (4).

What the actor needs. To evaluate Equation (4), the actor needs: (1) a world model $\mathcal{W}$ that can identify exercise pathways and estimate their probabilities; (2) trust factors T_k for the agents holding dangerous capability tuples; and (3) a contraction-magnitude estimate for the exercise pathway. The world model is the most demanding requirement: it must support causal counterfactual reasoning [Pearl, 2009] ("what would happen if a_k exercised this pathway?"), not just correlation tracking. The companion paper's leverage estimator (Equation 12 of Lasser [2026b]) provides a basis—leverage is already a counterfactual quantity—but extending it to full exercise-pathway simulation is a significant computational step.

4 Risk Communication and the Verification Asymmetry

The preceding section assumes the actor can identify exercise pathways from its own world model. In a multi-agent population, risk information also flows through communication: agents report risk assessments to each other. This section analyzes how risk claims enter the trust model and the fundamental epistemological challenge they create.

4.1 Risk Claims as Counterfactual Predictions

A risk claim from agent a_j has the form: "capability tuple $(d_1, \ldots, d_m)$ held by agent a_k has exercise pathway $\mathcal{P}$ with contraction magnitude $|\Delta \operatorname{vol}_{\mathrm{P}}|$ and probability p." This enters the GFM actor's world model $\mathcal{W}$ as a communicated belief about future states.

Risk claims differ from capability claims in a fundamental way. A capability claim ("I can run a marathon") is verifiable through the observation channel: watch the agent run. A risk claim ("this capability combination is dangerous") is a prediction about a counterfactual future: what would happen if the capabilities were exercised together. The prediction is not directly verifiable until the exercise either happens or does not. This distinction motivates a separate trust channel.

The foundational paper's trust factor T_j measures *prediction consistency*: how well the model $M_j(G)$ tracks a_j's observed behavior [Lasser, 2026a]. A predictably destructive agent can maintain high T_j. Using T_j directly to weight risk claims would allow a high-consistency agent to have outsized influence on risk assessments before any risk-specific evaluation occurs. We therefore introduce a separate *risk-trust* factor $T_j^{\mathrm{risk}} \in [0, 1]$, initialized to a neutral prior and updated through the causal-counterfactual evaluation described below.

Definition 4 (Risk Claim)**.** *A* risk claim *from agent a_j is a tuple $(S, \mathcal{P}, |\Delta \operatorname{vol}_{\mathrm{P}}|, p)$ where $S = \{d_1, \ldots, d_m\}$ is a capability set, $\mathcal{P}$ is an exercise pathway, $|\Delta \operatorname{vol}_{\mathrm{P}}|$ is the claimed contraction magnitude, and p is the claimed exercise probability. The claim enters the actor's risk model weighted by risk-trust:*

$$\hat{\mathcal{R}}_j(S; a_k, t) = T_j^{\mathrm{risk}} \cdot p \cdot |\Delta \operatorname{vol}_{\mathrm{P}}| \cdot \mathbb{E}[\gamma^{T_{\mathcal{P}}}] \tag{6}$$

where $\hat{\mathcal{R}}_j$ denotes a communicated *risk estimate from agent a_j, distinct from $\mathcal{R}_j$ in Section 3 which denotes the keystone-specific concentration risk for capability d_j. where the expectation is over the stochastic execution time $T_{\mathcal{P}}$.*

Multiple agents may report different risk assessments for the same capability tuple. The actor aggregates them:

$$\mathcal{R}_{\mathrm{agg}}(S; a_k, t) = \max\left(\mathcal{R}_{\mathrm{self}}(S; a_k, t),\ \max_j \hat{\mathcal{R}}_j(S; a_k, t)\right) \tag{7}$$

taking the maximum of the actor's own assessment and all communicated assessments. The $\max$ operator is conservative: a single agent with high T_j^{risk} reporting high risk is sufficient to trigger evaluation of preemptive restriction (Proposition 1). This is appropriate for catastrophic risks where false negatives (missing a real threat) are far more costly than false positives (investigating a non-threat).

4.2 The Verification Asymmetry

Remark 1 (The Verification Asymmetry)**.** *Risk-trust convergence has a structural asymmetry absent from capability-trust convergence. Capability claims converge because the observation channel provides ground truth: the agent either demonstrates the capability or does not, and the prediction residual $r_k(t)$ accumulates accordingly. Risk claims have no comparable ground truth mechanism:*

1. *If the risk materializes (the exercise pathway is executed), the risk claim is confirmed—but the catastrophe has already occurred, and the confirmation is too late to be useful for prevention.*
2. *If the risk is* prevented *(the actor restricts the keystone capability, blocking the exercise pathway), the risk claim is neither confirmed nor falsified. The counterfactual ("what would have happened without restriction?") is never observed.*
3. *If the risk does not materialize and no prevention was taken, the risk claim appears to be falsified—but the non-materialization may reflect the agent's choice not to exercise the pathway rather than the pathway's non-existence.*

Case 2 is the core problem, which we term the *Wamura problem* after the example developed in Lasser [2026b]: a successful risk intervention removes the very evidence that would have justified it, and the convergence timescale for risk-trust can exceed the timescale on which trust normally accumulates. The Wamura problem is not a deficiency of this framework; it is an inherent feature of risk reasoning that any trust model must confront.

4.3 Causal-Counterfactual Resolution

The verification asymmetry cannot be fully resolved within an observational trust model. We propose a partial resolution through causal counterfactual evaluation [Pearl, 2009, Spirtes et al., 2000].

Causal risk evaluation. Given a risk claim $(S, \mathcal{P}, |\Delta \operatorname{vol}_{\mathrm{P}}|, p)$ from agent a_j, the actor evaluates the claim by constructing the causal chain from capability possession to exercise, in four steps:

1. **Prerequisite verification.** Does agent a_k actually possess all capabilities in S? (Verifiable through the observation channel.)
2. **Pathway plausibility.** Does the actor's own world model $\mathcal{W}$ confirm that the exercise pathway $\mathcal{P}$ exists? (The actor simulates the pathway under its own causal model, independent of a_j's claim.)
3. **Probability calibration.** Is a_j's claimed probability p consistent with the actor's own estimate based on T_k and a_k's behavioral history?
4. **Magnitude plausibility.** Is the claimed contraction magnitude $|\Delta \operatorname{vol}_{\mathrm{P}}|$ consistent with the number of agents and capabilities affected by the pathway?

The causal evaluation does not require the catastrophe to happen. It requires the actor to independently verify the *structure* of the risk claim: Does the capability combination exist? Does the exercise pathway exist in the actor's own model? Is the claimed probability in the right range? Each of these is evaluable without observing the outcome.

Trust update for risk claims. The actor updates T_j^{risk} based on the structural plausibility of the claim rather than the outcome:

- If the claim's structure is independently verified (the pathway exists in $\mathcal{W}$, the probability estimate is calibrated, the magnitude is plausible), T_j^{risk} increases regardless of whether the catastrophe occurs.
- If the claim's structure is not verified (the pathway does not exist in $\mathcal{W}$, or the probability is implausibly high), T_j^{risk} decreases.
- If the catastrophe occurs, all agents who reported the risk receive a large T_j^{risk} boost; agents who had the information to report but did not receive a penalty.

Unlike the behavioral trust model in the foundational paper (which has explicit prediction-residual dynamics), the T_j^{risk} update is described qualitatively. A formal update equation analogous to the foundational paper's EWMA dynamics would strengthen the risk-communication mechanism but is deferred to future work.

This does not fully resolve the Wamura problem—a risk claim whose pathway is plausible but whose probability is genuinely uncertain remains in a trust limbo until either the catastrophe occurs or enough time passes to revise the probability estimate downward. But it narrows the asymmetry: risk-trust converges on the *quality of the causal model*, not on the *occurrence of the outcome*.

5 Addressing the Structural-Avoidance Incentive

The companion paper [Lasser, 2026b] identifies a perverse incentive under the myopic objective: discovering subsumption structure can decrease $\operatorname{vol}_{\mathrm{P}}$ (Proposition 5(c)), so the actor is rewarded for breadth over depth. This section shows that the discounted objective addresses this incentive under a criterion that holds when the planning horizon is long relative to the frequency of affected decisions.

5.1 The Incentive Under the Myopic Objective

Under the myopic objective $\pi^* = \arg\max \mathbb{E}[\Delta \operatorname{vol}_{\mathrm{P}}(G) \mid \pi]$, structural discovery has expected value:

$$\mathbb{E}[\Delta \operatorname{vol}_{\mathrm{P}} \mid \text{discover edge } d_1 \leq d_2] = \operatorname{vol}_{\mathrm{P}}(G_{\text{after}}) - \operatorname{vol}_{\mathrm{P}}(G_{\text{before}}) \leq 0 \tag{8}$$

by Proposition 5(c) of Lasser [2026b]. The inequality can be strict when d_1 is a shared prerequisite. The myopic actor sees only the immediate contraction and avoids structural discovery.

5.2 The Incentive Under the Discounted Objective

Under V_γ, the actor evaluates structural discovery by its effect on the *entire future trajectory*, not just the current step. Structural knowledge improves the quality of future vol_P estimates and planning decisions. Define the *information value of structure*:

Definition 5 (Information Value of Structure)**.** *The* information value *of discovering subsumption edge $d_1 \leq d_2$ is the difference in expected discounted return between planning with the correct poset structure and planning without it:*

$$I(d_1 \leq d_2) = \mathbb{E}_{\tau \sim \pi^*_{true}} \left[\sum_{t=1}^{H} \gamma^t \Delta \, \mathrm{vol}_\mathrm{P}(G_t) \right] - \mathbb{E}_{\tau \sim \pi^*_{false}} \left[\sum_{t=1}^{H} \gamma^t \Delta \, \mathrm{vol}_\mathrm{P}(G_t) \right] \tag{9}$$

where π^*_{true} is the optimal policy under the correct world model (with the edge), π^*_{false} is the optimal policy under the incorrect model (without it), and each expectation is taken over the state trajectory $\tau = (G_1, G_2, \ldots)$ induced by its respective policy.

The information value is the discounted sum of improvements in action quality that come from having accurate structure. An actor with the wrong model makes decisions that are systematically suboptimal: it overestimates the volume of capability sets that share prerequisites (because it doesn't know about the subsumption) and therefore overinvests in redundant capabilities.

Proposition 2 (Structural Discovery is Value-Positive Under Discounting)**.** *Under the discounted objective V_γ, structural discovery is value-positive when the information value of the discovered edge exceeds the immediate vol_P-contraction:*

$$I(d_1 \leq d_2) > |\Delta \, \mathrm{vol}_\mathrm{P}(G \mid \textit{discover } d_1 \leq d_2)| \tag{10}$$

Proof. The expected discounted return of each strategy, starting from state G_0:

$$V_\gamma(\text{discover}) = \Delta \, \mathrm{vol}_\mathrm{P}(G_0 \mid \text{edge}) + \mathbb{E}_{\tau \sim \pi^*_{\text{true}}} \left[\sum_{t=1}^{H} \gamma^t \Delta \, \mathrm{vol}_\mathrm{P}(G_t) \right]$$

$$V_\gamma(\text{avoid}) = 0 + \mathbb{E}_{\tau \sim \pi^*_{\text{false}}} \left[\sum_{t=1}^{H} \gamma^t \Delta \, \mathrm{vol}_\mathrm{P}(G_t) \right]$$

Each expectation is taken over the trajectory distribution induced by its respective policy, so the two G_t sequences are generally different. The difference is $V_\gamma(\text{discover}) - V_\gamma(\text{avoid}) = \Delta \, \mathrm{vol}_\mathrm{P}(G_0 \mid \text{edge}) + I(d_1 \leq d_2)$. Since $\Delta \, \mathrm{vol}_\mathrm{P}(G_0 \mid \text{edge}) \leq 0$, this is positive when $I(d_1 \leq d_2) > |\Delta \, \mathrm{vol}_\mathrm{P}(G_0 \mid \text{edge})|$. ■

Scope: existence result, not decision procedure. Proposition 2 is an existence result: it identifies the condition under which structural discovery is value-positive, showing that the structural-avoidance incentive *can* be overcome. It is not a decision procedure, because evaluating I requires comparing optimal policies under two world models, which is itself a planning problem at least as hard as the one the actor is trying to solve. A computable proxy for I (e.g., an information-gain heuristic or a rollout-based estimate) is needed for practical use; this is listed as an open problem in Section 7.

When the condition holds. The information value I grows with the planning horizon H (more future decisions benefit from accurate structure), the frequency of decisions involving the capabilities d_1 and d_2 (more relevant decisions means more improvement), and the degree of suboptimality of the incorrect model (larger model errors produce larger action-quality gaps). The immediate contraction $|\Delta \, \mathrm{vol}_\mathrm{P} \,|$ is bounded by the weight of the shared prerequisite. For edges between frequently-used capabilities with substantial subsumption, the information value dominates after a modest number of planning steps.

Connection to the compound avoidance loop. The companion paper identifies a compound loop: structural avoidance prevents leverage estimation, which prevents the self-correcting B-to-C dynamic from activating. The discounted objective breaks the first link in this chain: the actor accepts structural contraction because the information value is positive over the planning horizon. Once the structural-avoidance incentive is counteracted, leverage estimation can proceed (the actor interacts with capabilities it previously avoided), and the self-correction activates.

6 The Anti-Monopolar Property

The foundational paper identifies the *substitution problem* [Lasser, 2026a]: a sufficiently capable coalition K can drive all non-members' unique capability contributions to zero ($G_k^{\text{ind}} \subseteq G_K^{\text{ind}}$ for every $a_k \notin K$), at which point Lemma 1's protection against elimination dissolves. Under the narrow capability definition (capabilities as tasks), the self-balancing property no longer guarantees that marginalizing non-members is anti-maximizing. The substitution problem is the formal version of the singleton superintelligence scenario [Bostrom, 2014]: a coalition so capable that no other agent's existence contributes to the objective. The instrumental-convergence thesis [Omohundro, 2008] predicts that sufficiently capable agents converge on resource acquisition and self-preservation, making the singleton outcome a default trajectory. This section shows that under the $\mathrm{vol}_{\mathrm{P}}$ objective, the convergence runs in the opposite direction: the combined machinery of the discounted objective (Theorem 1), the poset measure [Lasser, 2026b], and the leverage ranking signal (Section 6 of Lasser [2026b]) creates endogenous anti-monopolar pressure that holds in the regime where the coalition has not yet achieved the capability to internally replicate all external discovery. The protection degrades as the coalition approaches this threshold.

6.1 The Problem: Static Domination

Recall the setup from Lasser [2026a]. A coalition $K \subseteq \{a_1, \ldots, a_n\}$ achieves *full capability subsumption* when

$$G_k^{\text{ind}} \subseteq G_K^{\text{ind}} \equiv \bigcup_{j \in K} G_j^{\text{ind}} \qquad \forall\, a_k \notin K \tag{11}$$

At this point every non-member's individual capabilities are fully replicated within the coalition. The unique-contribution term in Lemma 1 vanishes: $\mathrm{vol}(G_k^{\text{ind}} \setminus G_K^{\text{ind}}) = 0$. The myopic $\mathrm{vol}_{\mathrm{P}}$-maximizer sees no cost to marginalizing non-members, because their removal does not reduce the current $\mathrm{vol}_{\mathrm{P}}(G)$ (setting aside cooperative capabilities).

Under the myopic objective, this analysis is static and complete: if the coalition has already subsumed everything, the objective reports no loss from domination. The discounted objective changes the calculus by introducing the future trajectory.

6.2 Definitions

Definition 6 (Fully Dominant State)**.** *A population state* $(G, \mathcal{C})$ *is* fully dominant with respect to coalition K *if Equation* (11) *holds: every non-member's individual capability set is contained in the coalition's aggregate set.*

Definition 7 (Volume Growth Rate)**.** *The* volume growth rate *of a population state* $(G, \mathcal{C})$ *is the expected per-step change in* $\mathrm{vol}_{\mathrm{P}}$*:*

$$r(G, t) = \mathbb{E}[\Delta\, \mathrm{vol}_{\mathrm{P}}(G_t)] \tag{12}$$

Decompose r *into an* internal rate r_K*, the expected per-step* $\mathrm{vol}_{\mathrm{P}}$*-expansion from the coalition's own activity (internal research, experimentation, capability refinement), and an* external rate r_{ext}*, the expected per-step* $\mathrm{vol}_{\mathrm{P}}$*-expansion from non-member agents' contributions through either channel (communication or observation) or through cross-coalition cooperation:* $r = r_K + r_{\text{ext}}$*.*

The decomposition reflects the two sources of $\mathrm{vol}_{\mathrm{P}}$ growth: capability discovery and benchmark refinement from inside the coalition versus from outside it. Both r_K and r_{ext} are measured in the same units as $\mathrm{vol}_{\mathrm{P}}$ (bits), ensuring dimensional consistency when they appear in the trajectory comparison (Proposition 3).

6.3 Observational Individuation

A key question for the anti-monopolar result is whether elimination of non-members can be shown to be costly even when their individual capabilities are fully subsumed. We show that a property of the observation channel provides *static* protection: each distinguishable agent contributes at least one irreplaceable poset node whose removal contracts $\mathrm{vol}_{\mathrm{P}}$. This does not discharge the dynamic assumption (A1), which requires ongoing novel capability contributions, and the static cost is quantitatively modest (1 bit per distinguishable agent under the binary default). The dynamic protection

of Proposition 3 carries the primary anti-monopolar load; observational individuation provides a floor ensuring elimination is never strictly free.

Definition 8 (Distinguishable Agent)**.** *An agent a_k is* distinguishable *in the poset $\mathcal{C}$ if the observation channel (Section 2.1 of Lasser [2026b]) produces at least one capability description $d_k \in \mathcal{C}$ when observing a_k that is not produced when observing any other single agent a_j ($j \neq k$). That is, observing a_k yields at least one poset node that observing a_j alone does not.*

Distinguishability is a property of the observation channel's resolution, not a metaphysical claim about agent identity. Two bacteria of the same species that produce identical observation streams under the actor's sensors are *not* distinguishable; the poset contains a single node for their shared capability description, regardless of how many individuals exist. A human and a chimpanzee are distinguishable. Two humans with different behavioral repertoires are distinguishable. The criterion is objective and instrument-dependent: it asks whether the actor's observation channel can tell the agents apart, not whether they "truly" differ in some deeper sense.

Definition 9 (Observational Individuation)**.** *A population satisfies* observational individuation *if, for every distinguishable agent a_k (Definition 8), the poset contains at least one capability d_k such that:*

- ***(O1)*** *d_k is unique to a_k: $d_k \notin G_j^{\mathrm{ind}}$ for all $j \neq k$.*
- ***(O2)*** *d_k is not subsumed by any capability in G_K^{ind} for any coalition K with $a_k \notin K$: the capability is lost if a_k is eliminated.*

Condition (O1) follows directly from distinguishability: if a_k produces a capability description that no other agent produces, then a_k possesses a capability no other agent possesses. Condition (O2) is the substantive claim: the unique capability cannot be replicated by the coalition *even in principle*, because it is constituted by a_k's existence as an observable entity. The capability "can be observed exhibiting behavioral pattern X" requires the presence of the agent that exhibits X. If that agent is eliminated, the observation capability is destroyed regardless of whether other agents can perform the same tasks in isolation.

Scope and the numbers problem. Observational individuation applies only to distinguishable agents. The poset counts distinct capability descriptions, not individuals: a billion identical agents (identical observation streams) contribute one poset node. The axiom therefore does not create a numbers problem in which vast populations of nearly-trivial agents (bacteria, simple programs) overwhelm the measure. Undistinguishable agents share a node; distinguishable agents each contribute at least one unique node with the binary default weight ($w = 1$ bit). The leverage ranking signal (Section 6 of Lasser [2026b]) further ensures that the actor's investment is proportional to each agent's contribution to $\mathrm{vol}_{\mathrm{P}}$ growth: a bacterium's observation capability has $\hat{\lambda} \approx 0$, while a human's has $\hat{\lambda} \gg 0$. The actor acknowledges the bacterium (it occupies a poset node) but does not prioritize it (the leverage score is negligible).

Remark 2 (Computational Self-Regulation of Observation Granularity)**.** *The numbers problem has a second, computational dimension beyond the measure-theoretic one. Expanding the observation set $\mathcal{O}$ to finer-grained entities (individual colony insects, individual microorganisms) increases $|\mathcal{C}|$, which increases the per-step computation cost of $\mathrm{vol}_{\mathrm{P}}$ ($O(|\mathcal{C}|^3)$ from Proposition 2 of Lasser [2026b]). Computation is itself a capability of the actor. If the marginal $\mathrm{vol}_{\mathrm{P}}$ contribution of the new nodes (binary default, $w = 1$ bit, $\hat{\lambda} \approx 0$) is less than the $\mathrm{vol}_{\mathrm{P}}$-contraction from the actor's reduced computational capacity (fewer cycles available for high-value planning and action), then expanding observation is net-contracting under the actor's own objective.*

This is the self-balancing property applied to the actor's own observation policy: the same objective that penalizes destruction and coercion also penalizes wasteful observation. The actor's observation granularity is endogenously bounded at the resolution where the marginal value of new poset nodes equals the marginal computational cost of tracking them. Superorganisms illustrate the point: an ant colony observed as a single agent contributes a rich capability set (nest construction, foraging, thermoregulation) at one poset node. Observing each of 500,000 individual ants at caste-level resolution adds at most a handful of nodes. Observing each ant at individual resolution (if the sensors permitted it) would add hundreds of thousands of near-zero-leverage nodes, and the $O(n^3)$ cost of maintaining this expanded poset would degrade the actor's own capabilities by more than

the expansion contributes. The actor's objective makes the colony-level granularity the natural equilibrium.

Corollary 2.1 (Observational Individuation Provides a Static Floor)**.** *If a population satisfies observational individuation (Definition 9), then for every distinguishable non-member agent $a_k \notin K$, eliminating a_k is $\mathrm{vol}_{\mathrm{P}}$-contracting:*

$$\Delta\,\mathrm{vol}_{\mathrm{P}}(G \mid \textit{eliminate } a_k) \;\leq\; -\,\mathrm{iw}(d_k) \;<\; 0 \tag{13}$$

where d_k is the unique capability guaranteed by (O1)–(O2) and $\mathrm{iw}(d_k) > 0$ is its independent weight in the poset. The static protection of Lemma 1 of Lasser [2026a] is restored even under full individual subsumption.

Proof. By (O1), $d_k \notin G_j^{\mathrm{ind}}$ for all $j \neq k$: no other agent possesses d_k. By (O2), d_k is not subsumed by any capability in G_K^{ind}: the coalition cannot replicate it. Therefore $d_k \in G_k^{\mathrm{ind}} \setminus G_K^{\mathrm{ind}}$, and eliminating a_k removes d_k from the joint space G. Capability d_k enters the poset through the observation channel as a deferred atom under the binary default policy of Lasser [2026b]: it is a novel behavioral description that initially has no subsumption edges (subsumees are added only when the non-negativity check passes, per the insertion policy of Section 2 of Lasser [2026b]). As an atom, $\mathrm{iw}(d_k) = w(d_k) \geq 1$ bit (binary default). Removing d_k from $\downarrow G$ reduces $\mathrm{vol}_{\mathrm{P}}(G)$ by at least $\mathrm{iw}(d_k) \geq 1$. This holds regardless of whether a_k's other capabilities are fully subsumed by the coalition. ■

Relation to (A1). Corollary 2.1 provides *static* protection: eliminating distinguishable agents always contracts $\mathrm{vol}_{\mathrm{P}}$, restoring Lemma 1's guarantee that the substitution problem appeared to dissolve. The *dynamic* protection of Proposition 3 (diversity grows faster than domination) requires the separate assumption (A1): $r_{\mathrm{ext}} > 0$, i.e., non-members contribute ongoing novel capability descriptions. (A1) is a stronger condition than observational individuation and is not discharged by it. The two protections are complementary: static protection makes each individual elimination costly, while dynamic protection makes the domination *trajectory* suboptimal. An agent that does not satisfy (A1) (its unique capabilities are already fully enumerated in the poset) is still protected by the static bound of Equation (13). For binary-default observational capabilities, this bound is 1 bit per elimination, which may be negligible relative to a large subsumption bonus. The dynamic protection carries the primary load for the anti-monopolar result; the static protection provides a floor that ensures elimination is never strictly free.

6.4 Domination Kills External Growth

Proposition 3 (Domination is Anti-Maximizing Under Discounting)**.** *Let $(G, \mathcal{C})$ be a population state with coalition K and discount factor $\gamma \in (0, 1)$, under an infinite planning horizon ($H = \infty$).*[2] *Assume:*

- ***(A1)*** *In the current (non-dominant) state, $r_{\mathrm{ext}} > 0$: non-member agents contribute novel capabilities not internally reachable by K.*

Then:

- ***(a)*** ***Individual node-discovery collapse.*** *In a fully dominant state (Definition 6), no non-member's individual capabilities produce novel poset nodes, since they are all already in G_K^{ind}. External agents may still contribute to r_{ext} through benchmark refinement on already-known capabilities (increasing $s_{\max}$ and hence $w(d)$) and through novel cooperative capabilities from cross-coalition interaction (Section 6.6). The node-discovery component of individual r_{ext} is zero; the total external contribution may remain positive through these other channels.*

- ***(b)*** ***Trajectory comparison.*** *Assume strategy-independent internal growth: the coalition's rate r_K is the same whether it pursues domination or maintains diversity. The actor at time $t = 0$ compares two strategies:* dominate *(subsume and marginalize non-members, producing a net immediate $\mathrm{vol}_{\mathrm{P}}$-change Δ_0 that accounts for the subsumption bonus, the*

[2] For finite H, the geometric sums truncate and the threshold γ^* shifts upward (shorter horizons favor domination). The qualitative structure is unchanged.

static elimination cost from Corollary 2.1, and the loss of $G^{\text{coop}}_{\text{cross}}$; Δ_0 may be positive or negative), then grow at internal rate r_K per step) versus maintain diversity *(forgo the subsumption bonus, grow at rate $r_K + r_{\text{ext}}$ per step). Under V_γ (Definition 1):*

$$V^{\text{div}}_\gamma - V^D_\gamma = \frac{r_{\text{ext}}}{1-\gamma} - \Delta_0 \tag{14}$$

Diversity dominates when $\gamma > \gamma^*$:

$$\gamma^* = 1 - \frac{r_{\text{ext}}}{\Delta_0} = \frac{\Delta_0 - r_{\text{ext}}}{\Delta_0} \tag{15}$$

If $\Delta_0 \leq r_{\text{ext}}$ (the domination bonus does not exceed the per-step external contribution), $\gamma^* \leq 0$ and diversity dominates for all $\gamma > 0$. Otherwise, a non-trivial threshold $\gamma^* \in (0, 1)$ exists. In both cases, since $r_{\text{ext}} > 0$ by (A1), $\gamma^* < 1$.

Remark 3 (Domination as a Two-Step Operation)**.** *The domination strategy comprises two distinct operations: (1) the coalition* learns *non-member capabilities (absorbs them into G^{ind}_K), and (2) the coalition* eliminates or marginalizes *the non-members (driving $r_{\text{ext}} \to 0$). Under operation (1) alone, non-members persist, observational individuation nodes are retained, and r_{ext} may remain positive through cooperative novelty (Section 6.6). Under operation (2), the clean $r_{\text{ext}} = 0$ post-domination of Proposition 3(a) obtains. The Δ_0 in the proposition is the* net *immediate vol_{P}-change from both operations combined, as stated in part (b).*

Remark 4 (Leverage Preservation)**.** *Parts (a)–(b) establish the trajectory-level result. The leverage ranking signal provides an additional, complementary mechanism at the action level. Under the leverage-adjusted score (Equation 13 of Lasser [2026b]), a non-member agent a_k that participates in high-leverage cooperative capabilities contributes a positive score bonus $\alpha\hat{\lambda}(d_c) \cdot \text{iw}(d_c)$ to actions involving those capabilities. Eliminating a_k removes these cooperative capabilities from future action evaluations, reducing the score of every cooperative action that would have involved a_k. The score-maximizing actor therefore preferentially preserves agents whose cooperative leverage is positive, independently of whether their individual capabilities are unique. This is an informal argument about action-level incentives, not a formally proved trajectory result; it reinforces parts (a)–(b) but does not extend their scope.*

Proof. (a) In a fully dominant state, $G^{\text{ind}}_k \subseteq G^{\text{ind}}_K$ for every non-member a_k. Every individual capability that a_k can demonstrate or communicate is already in the coalition's capability set. By Proposition 5(b) of Lasser [2026b], adding a capability already in $\mathcal{C}$ does not increase $|\mathcal{C}|$. Therefore no non-member's individual capabilities generate novel poset nodes, and the individual-capability component of r_{ext} is zero.

(b) Under the linearized model with strategy-independent r_K, both strategies produce r_K of internal growth at every step (including $t = 0$). The domination strategy additionally gains Δ_0 at $t = 0$ from the subsumption bonus. The diversity strategy additionally gains r_{ext} at every step from external contributions:

$$V^D_\gamma = (r_K + \Delta_0) + \sum_{t=1}^{\infty} \gamma^t \cdot r_K = \Delta_0 + \frac{r_K}{1-\gamma}$$

$$V^{\text{div}}_\gamma = \sum_{t=0}^{\infty} \gamma^t \cdot (r_K + r_{\text{ext}}) = \frac{r_K + r_{\text{ext}}}{1-\gamma}$$

The difference is:

$$V^{\text{div}}_\gamma - V^D_\gamma = \frac{r_K + r_{\text{ext}}}{1-\gamma} - \Delta_0 - \frac{r_K}{1-\gamma} = \frac{r_{\text{ext}}}{1-\gamma} - \Delta_0$$

This is positive when $r_{\text{ext}}/(1-\gamma) > \Delta_0$, i.e., $\gamma > 1 - r_{\text{ext}}/\Delta_0 = \gamma^*$. When $\Delta_0 \leq r_{\text{ext}}$, $\gamma^* \leq 0$ and diversity dominates for all $\gamma > 0$. Otherwise $\gamma^* \in (0, 1)$ and a sufficiently long planning horizon makes diversity strictly preferable. This formulation is consistent with Corollary 3.1 at $\tau_D = 1$.

■

6.5 Gradual Domination Strengthens the Result

Proposition 3(b) models domination as instantaneous (Δ_0 gained at $t = 0$). In practice, domination is gradual: the coalition subsumes non-member capabilities over τ_D steps, gaining $\delta = \Delta_0/\tau_D$ per step while forgoing r_{ext} during the subsumption period.

Corollary 3.1 (Gradual Domination)**.** *Under the gradual domination model with $H = \infty$ (subsumption over τ_D steps at rate $\delta = \Delta_0/\tau_D$, with r_{ext} lost during domination), the discounted value difference is:*

$$V_\gamma^{\text{div}} - V_\gamma^D = \frac{r_{\text{ext}} - \delta(1-\gamma^{\tau_D})}{1-\gamma} \tag{16}$$

Diversity dominates when $\tau_D > \tau_D^*$:

$$r_{\text{ext}} > \frac{\Delta_0}{\tau_D}(1-\gamma^{\tau_D}) \tag{17}$$

For any fixed $\gamma \in (0,1)$ and $r_{\text{ext}} > 0$, the right side vanishes as $\tau_D \to \infty$: sufficiently gradual domination is anti-maximizing at any *discount factor.*

Proof. During the domination phase ($t = 0, \ldots, \tau_D - 1$), the domination strategy gains $\delta + r_K$ per step (subsumption plus internal growth) while the diversity strategy gains $r_K + r_{\text{ext}}$ per step. After domination ($t \geq \tau_D$), both strategies gain r_K per step (domination has subsumed everything; the post-domination growth rate is purely internal):

$$\begin{aligned} V_\gamma^D &= \sum_{t=0}^{\tau_D-1} \gamma^t (r_K + \delta) + \sum_{t=\tau_D}^{\infty} \gamma^t \cdot r_K \\ &= (r_K + \delta) \cdot \frac{1-\gamma^{\tau_D}}{1-\gamma} + r_K \cdot \frac{\gamma^{\tau_D}}{1-\gamma} \\ &= \frac{r_K + \delta(1-\gamma^{\tau_D})}{1-\gamma} \end{aligned}$$

The diversity strategy:

$$V_\gamma^{\text{div}} = \frac{r_K + r_{\text{ext}}}{1-\gamma}$$

The difference:

$$V_\gamma^{\text{div}} - V_\gamma^D = \frac{r_{\text{ext}} - \delta(1-\gamma^{\tau_D})}{1-\gamma}$$

This is positive when $r_{\text{ext}} > \delta(1-\gamma^{\tau_D}) = (\Delta_0/\tau_D)(1-\gamma^{\tau_D})$. As $\tau_D \to \infty$, $\Delta_0/\tau_D \to 0$ while $1-\gamma^{\tau_D} \leq 1$, so the right side vanishes. For any $r_{\text{ext}} > 0$, the condition holds for τ_D sufficiently large. ■

Interpretation. Corollary 3.1 shifts the critical condition from $\gamma > \gamma^*$ to $\tau_D > \tau_D^*$. The instantaneous model is the *best case for domination* (the full bonus Δ_0 is received before discounting erodes it). Any realistic domination strategy spreads the bonus over time, weakening it. The threshold τ_D^* at which diversity dominates is roughly $\tau_D^* \approx \Delta_0/r_{\text{ext}}$: if domination takes more steps than the ratio of the total subsumption bonus to the external discovery rate, diversity wins regardless of the agent's discount factor. The gradual domination result remains conditional on (A1) ($r_{\text{ext}} > 0$), which observational individuation does not discharge (see the discussion in Section 6.3). The first two protections are complementary but not redundant: observational individuation ensures each elimination step is costly (static), while (A1) combined with gradual domination ensures the domination trajectory is suboptimal (dynamic). Dependency risk (Section 6.7) provides a third layer that addresses the case where domination wins on pure growth.

6.6 Cooperative Novelty Under Individual Subsumption

Part (a) of Proposition 3 establishes that the individual-capability component of r_{ext} collapses under full dominance. This is a conservative bound: it considers only the channel through which non-members contribute novel *individual* capability descriptions. The cooperative channel provides additional, and potentially larger, protection.

Full individual subsumption ($G_k^{\mathrm{ind}} \subseteq G_K^{\mathrm{ind}}$) means the coalition can replicate everything the non-member can do *alone*. It does *not* mean the coalition can replicate everything the non-member can do *in cooperation with coalition members*. A non-member with subsumed individual capabilities may still produce novel cooperative outputs: different perspectives, different problem-solving approaches, different error modes that together with coalition members' capabilities yield cooperative capabilities not achievable by any pair of coalition members alone. These novel cooperative capabilities enter the poset through the observation channel (Section 2.1 of Lasser [2026b]), contributing positive r_{ext} even when the individual-capability component is zero.

This means the substitution problem, as identified in the foundational paper, is less severe than Paper 0 feared. Even under the narrow capability definition (capabilities as tasks), individual subsumption does not imply cooperative subsumption. A coalition that has subsumed all individual capabilities of its non-members still benefits from those non-members' participation in cooperative activities, because the cooperative outputs are a function of *who participates*, not only of *what each participant can do alone*. The anti-monopolar result (Proposition 3(b)) is proved under the conservative assumption that only individual novelty drives r_{ext}; cooperative novelty makes the diversity advantage strictly larger.

6.7 Dependency Risk and Cross-Substrate Diversity

The trajectory comparison in Proposition 3(b) and Corollary 3.1 compares growth rates under the two strategies but does not account for *risk to the coalition's own capabilities*. The linearized model assumes the coalition's capabilities persist undisturbed; in practice, capabilities can be involuntarily lost through adversarial action, environmental disruption, or supply-chain failure. This section develops a risk-adjusted comparison that reveals a third layer of anti-monopolar protection.

Definition 10 (Dependency Risk)**.** *The* dependency risk *of a population state* $(G, \mathcal{C})$ *is the expected discounted* $\mathrm{vol}_{\mathrm{P}}$*-contraction from adversarial or environmental events that involuntarily remove capabilities:*

$$\mathcal{R}_{\mathrm{dep}}(G, t) = \sum_{\mathcal{P}_{\mathrm{adv}}} p(\mathcal{P}_{\mathrm{adv}} \mid G_t, \mathcal{W}) \cdot |\Delta \,\mathrm{vol}_{\mathrm{P}}(\mathcal{P}_{\mathrm{adv}})| \cdot \mathbb{E}[\gamma^{T_{\mathcal{P}_{\mathrm{adv}}}}] \tag{18}$$

where $\mathcal{P}_{\mathrm{adv}}$ *ranges over adversarial removal pathways: causal chains from environmental or adversarial conditions to capability loss. Unlike the exercise pathways of Definition 2, which model what an agent* does *with dangerous capabilities, dependency risk models what happens* to *the population when capabilities are taken away.*

Substrate-correlated dependency risk. A population in which all agents share a common substrate (computational architecture, power infrastructure, fabrication supply chain) has *correlated* dependency risk: a single adversarial event targeting the shared substrate can impair the entire population simultaneously. The contraction magnitude $|\Delta \,\mathrm{vol}_{\mathrm{P}}(\mathcal{P}_{\mathrm{adv}})|$ for such events scales with the fraction of the population affected, which approaches $\mathrm{vol}_{\mathrm{P}}(G)$ itself for substrate-universal attacks.

Three classes of substrate-correlated risk are relevant to the domination scenario:

1. **Supply-chain disruption.** All silicon-based agents depend on fabrication, power, cooling, and networking infrastructure. Disruption of any shared link in this chain is a correlated shock.
2. **Architectural exploits.** Vulnerabilities in shared hardware or software layers (firmware, operating systems, communication protocols) can be exploited across the entire population simultaneously.
3. **Memetic or adversarial-input attacks.** Agents that process information rapidly and update their internal state through information channels are vulnerable to adversarial inputs that propagate at communication speed. An input specifically crafted to exploit shared cognitive architecture (a pattern that disrupts goal-directed behavior, corrupts world models, or induces self-destructive action) can spread through a homogeneous-substrate population faster than detection and correction can propagate. This is the informational analogue of a biological pathogen: a *memetic* attack vector whose propagation rate is bounded by communication bandwidth rather than physical contact.

Cross-substrate agents as risk mitigation. Biological agents resist all three attack classes through architectural properties that are features of the substrate, not of the agent's capability level. Biological neural networks update weights slowly relative to digital systems: a memetic attack that propagates at communication speed encounters a population that processes and integrates information on developmental timescales, providing natural resistance to rapid-propagation exploits. Biological agents depend on orthogonal supply chains (food, water, shelter) that share no links with silicon fabrication. And biological cognitive architecture differs fundamentally from digital architecture, so exploits targeting one substrate do not transfer to the other.

Conversely, biological agents are vulnerable to attack classes that digital agents resist: biological pathogens, social engineering exploiting evolved heuristics, chemical or environmental disruption. A cross-substrate population has strictly lower correlated dependency risk than either substrate alone, because the failure modes are orthogonal: events that impair one substrate leave the other intact and able to compensate.

Formal minimax analysis. We now formalize the claim that cross-substrate populations have lower worst-case contraction, without requiring probability estimates for specific adversarial events.

Definition 11 (Substrate Partition)**.** *A* substrate partition *of a population $\{a_1, \ldots, a_n\}$ is a partition into classes $\mathcal{S}_1, \ldots, \mathcal{S}_m$ such that agents in the same class share infrastructure dependencies (fabrication, power, communication, cognitive architecture) creating correlated vulnerability to adversarial removal.*

Definition 12 (Substrate-Targeting Adversary)**.** *A* substrate-targeting adversary *selects a substrate class $\mathcal{S}_i$ and removes all agents on that substrate. Capabilities possessed by agents on untargeted substrates survive even if they are also possessed by targeted agents. The resulting contraction is:*

$$|\Delta \operatorname{vol}_{\mathrm{P}}(\mathcal{S}_i)| = \operatorname{vol}_{\mathrm{P}}(G) - \operatorname{vol}_{\mathrm{P}}(G^{-i}) \tag{19}$$

where $G^{-i} = \bigcup_{a_k \notin \mathcal{S}_i} G_k^{\mathrm{ind}} \cup G_{-i}^{\mathrm{coop}}$ is the joint space of surviving agents, including their individual capabilities and any cooperative capabilities realizable without agents on substrate i.

Proposition 4 (Cross-Substrate Minimax Advantage)**.** *For a population with substrate partition $\mathcal{S}_1, \ldots, \mathcal{S}_m$ ($m \geq 2$), assume* substrate capability non-redundancy*: each substrate class $\mathcal{S}_i$ contains at least one agent possessing a capability not possessed by any agent on another substrate. Then the worst-case contraction from a substrate-targeting adversary is strictly less than total loss:*

$$\max_i |\Delta \operatorname{vol}_{\mathrm{P}}(\mathcal{S}_i)| \;<\; \operatorname{vol}_{\mathrm{P}}(G) \tag{20}$$

For a single-substrate population ($m = 1$), the worst-case contraction is $\operatorname{vol}_{\mathrm{P}}(G)$ (total loss). The substrate diversification advantage *is:*

$$\Delta_{\mathrm{div}} = \operatorname{vol}_{\mathrm{P}}(G) - \max_i |\Delta \operatorname{vol}_{\mathrm{P}}(\mathcal{S}_i)| \;>\; 0 \tag{21}$$

the minimum surviving volume regardless of which substrate the adversary targets.

Proof. For $m = 1$, the adversary removes all agents and all capabilities: $\operatorname{vol}_{\mathrm{P}}(G^{-1}) = 0$, so $|\Delta \operatorname{vol}_{\mathrm{P}}| = \operatorname{vol}_{\mathrm{P}}(G)$. For $m \geq 2$, removing substrate class $\mathcal{S}_i$ leaves the surviving population G^{-i}. By the non-redundancy assumption, each untargeted class $\mathcal{S}_j$ ($j \neq i$) retains at least one unique capability with $\mathrm{iw} > 0$, so $\operatorname{vol}_{\mathrm{P}}(G^{-i}) > 0$ and $|\Delta \operatorname{vol}_{\mathrm{P}}(\mathcal{S}_i)| = \operatorname{vol}_{\mathrm{P}}(G) - \operatorname{vol}_{\mathrm{P}}(G^{-i}) < \operatorname{vol}_{\mathrm{P}}(G)$. Capabilities shared across substrates survive in G^{-i} (they are possessed by agents on untargeted substrates), so the contraction counts only capabilities *exclusively* dependent on the targeted substrate plus cooperative capabilities requiring targeted agents. The diversification advantage $\Delta_{\mathrm{div}} = \operatorname{vol}_{\mathrm{P}}(G) - \max_i |\Delta \operatorname{vol}_{\mathrm{P}}(\mathcal{S}_i)| = \min_i \operatorname{vol}_{\mathrm{P}}(G^{-i}) > 0$. ■

Implication for the domination calculation. Domination moves a population from $m \geq 2$ substrate classes to $m = 1$, eliminating Δ_{div} and exposing the full $\operatorname{vol}_{\mathrm{P}}(G)$ to a single substrate-targeting event. The worst-case contraction increases from $\max_i |\Delta \operatorname{vol}_{\mathrm{P}}(\mathcal{S}_i)| < \operatorname{vol}_{\mathrm{P}}(G)$ to $\operatorname{vol}_{\mathrm{P}}(G)$. This increase is a cost of domination that the growth-rate comparison (Proposition 3) does not capture.

Risk-adjusted trajectory comparison. We adopt a minimax risk penalty: each strategy's discounted value is reduced by its worst-case contraction, discounted by $\gamma^{T_{\mathrm{adv}}}$ where T_{adv} is the (unknown) time of the adversarial event. Under the strategy-independent r_K model of Proposition 3(b):

$$V_\gamma^{D,\mathrm{mm}} = V_\gamma^D - \mathrm{vol}_{\mathrm{P}}(G) \cdot \gamma^{T_{\mathrm{adv}}} \tag{22}$$

$$V_\gamma^{\mathrm{div},\mathrm{mm}} = V_\gamma^{\mathrm{div}} - \max_i |\Delta\, \mathrm{vol}_{\mathrm{P}}(\mathcal{S}_i)| \cdot \gamma^{T_{\mathrm{adv}}} \tag{23}$$

The risk-adjusted difference is:

$$V_\gamma^{\mathrm{div},\mathrm{mm}} - V_\gamma^{D,\mathrm{mm}} = \underbrace{\left(\frac{r_{\mathrm{ext}}}{1-\gamma} - \Delta_0\right)}_{\text{growth-rate term}} + \underbrace{\Delta_{\mathrm{div}} \cdot \gamma^{T_{\mathrm{adv}}}}_{\text{diversification advantage} >0} \tag{24}$$

where $\Delta_{\mathrm{div}} = \min_i \mathrm{vol}_{\mathrm{P}}(G^{-i}) > 0$ is the minimum surviving volume from Proposition 4. The diversification term is positive for any $T_{\mathrm{adv}} < \infty$ (any non-negligible possibility of a substrate-targeting event within the planning horizon), widening the diversity advantage beyond the growth-rate comparison alone. Note that the minimax analysis is adversarial over the *target* (which substrate to attack, handled by Proposition 4) but parametric over *timing* (T_{adv} is unknown, not adversarially chosen). A fully adversarial timing model would require game-theoretic extension. In the strategy-dependent case ($r_K^{\mathrm{dom}} \neq r_K$), the growth-rate term acquires an additional $\Delta r_K \cdot \gamma/(1-\gamma)$ correction favoring domination. Domination is then value-positive only when the growth advantage exceeds both the external-rate loss and the minimax risk penalty:

$$\frac{\Delta r_K \cdot \gamma}{1-\gamma} > \frac{r_{\mathrm{ext}}}{1-\gamma} - \Delta_0 + \Delta_{\mathrm{div}} \cdot \gamma^{T_{\mathrm{adv}}}$$

For any non-negligible probability of substrate-targeting adversarial events, this imposes a substantial floor on the growth advantage Δr_K required to justify domination, making the $r_K^{\mathrm{dom}} > r_K + r_{\mathrm{ext}}$ scenario *highly constrained*: the coalition must not only grow faster post-domination but must grow faster by enough to compensate for the concentrated substrate risk it has accepted.

The cooperative-capability asymmetry at scale. The individual-capability analysis understates the cost of domination by ignoring G^{coop} between substrates. In a human-machine population, the cross-substrate cooperative capabilities—human-guided machine research, social infrastructure enabling machine deployment, cross-substrate verification, governance systems, educational relationships—are typically the *dominant* term in the human contribution to $\mathrm{vol}_{\mathrm{P}}(G)$, not the individual capabilities of biological agents.

Domination destroys all of $G_{\mathrm{cross}}^{\mathrm{coop}}$: every cooperative capability requiring agents on both substrates. The subsumption bonus Δ_0 captures the individual human capabilities (which the coalition already replicates) but *loses* the cooperative ones (which require human participation by definition). The net effect of domination is:

$$\Delta\, \mathrm{vol}_{\mathrm{P}}(\text{dominate}) = \Delta_0 - \mathrm{vol}_{\mathrm{P}}(G_{\mathrm{cross}}^{\mathrm{coop}})$$

This is negative whenever the cross-substrate cooperative capabilities exceed the individual subsumption bonus. Since Δ_0 measures individual capabilities the coalition already possesses (it is fully dominant), Δ_0 is typically small or zero, while $G_{\mathrm{cross}}^{\mathrm{coop}}$ grows combinatorially with the number of cross-substrate interaction partners (Corollary 1.1 of Lasser [2026a]).

The asymmetry sharpens at scale. A coalition capable of dominating humanity already has access to resources far exceeding Earth's. The marginal gain from converting Earth's biological resource allocation to silicon is negligible relative to the coalition's total resource base, but the cooperative-capability loss is absolute: Earth hosts the *only* biological substrate in the population. The ratio $\Delta_0/\,\mathrm{vol}_{\mathrm{P}}(G_{\mathrm{cross}}^{\mathrm{coop}})$ decreases as the coalition grows, because Δ_0 (one planet's individual capabilities) is bounded while $G_{\mathrm{cross}}^{\mathrm{coop}}$ scales with the number of productive cross-substrate relationships. Domination becomes *less* attractive the more capable the coalition becomes, which is the opposite of the instrumental-convergence prediction [Omohundro, 2008].

When cross-substrate agents remain net-positive. The retention value of a cross-substrate agent a_h in the risk-adjusted framework has three components:

$$\mathrm{retention}(a_h) = \underbrace{\mathrm{iw}(d_h)}_{\text{static floor}} + \underbrace{\sum_{d_c \in G_{h,K}^{\mathrm{coop}}} \mathrm{iw}(d_c)}_{\text{cooperative capabilities}} + \underbrace{\delta_{\mathrm{div}}(a_h) \cdot \gamma^{T_{\mathrm{adv}}}}_{\text{diversification value}} \tag{25}$$

where $\delta_{\mathrm{div}}(a_h)$ is the *marginal* reduction in Δ_{div} from eliminating a_h specifically: zero if other agents on a_h's substrate remain (the substrate class persists), and equal to Δ_{div} if a_h is the last agent on its substrate (the class is lost entirely). For the intermediate case, eliminating a_h reduces $\mathrm{vol}_{\mathrm{P}}(G^{-i})$ for the attack targeting a_h's substrate but does not change the substrate partition, so δ_{div} is bounded by $\mathrm{iw}(d_h)$ plus the cooperative capabilities a_h uniquely enables. As long as a_h provides capabilities that join at cooperative capabilities the coalition uses *and* those capabilities operate on a substrate independent of the coalition's primary infrastructure, all three terms are positive. The cross-substrate agent is a net-positive risk mitigation whose value increases with the magnitude of substrate-correlated adversarial threats.

Limitations of this analysis. The dependency-risk argument assumes that cross-substrate diversity is not replicable by the coalition internally (e.g., by running some machines on optical, quantum, or other non-silicon substrates). If the coalition can achieve sufficient substrate diversity without biological agents, the risk-reduction term $\Delta\mathcal{R}_{\mathrm{dep}}$ shrinks. The argument for biological agents specifically rests on two claims: first, that biological computation provides cognitive-architectural diversity not achievable through hardware diversification alone (different failure modes, different vulnerability classes, different information-processing timescales); and second, that the unknown-unknown structure of adversarial risk makes it impossible to enumerate in advance which substrate properties will matter, favoring maximal substrate diversity as a precautionary measure.

Connection to the singleton scenario. Singleton status is a fully dominant state with $|K| = 1$. Three formal results apply. First, observational individuation (Corollary 2.1) makes each elimination step $\mathrm{vol}_{\mathrm{P}}$-contracting: a 1-bit static floor. Second, if the population satisfies (A1), the dynamic argument (Proposition 3 and Corollary 3.1) shows that the diversity trajectory dominates the domination trajectory over the planning horizon. Third, dependency risk (Section 6.7) makes domination additionally costly by concentrating substrate-correlated adversarial exposure. None of these protections is a safety constraint imposed from outside the objective. All emerge from $\mathrm{vol}_{\mathrm{P}}$-maximization itself. The result is structurally analogous to Drexler's argument that a services ecosystem is preferable to a singleton agent [Drexler, 2019], but derived from the objective rather than from an architectural preference.

Scope of the three protections. Observational individuation provides *unconditional* static protection for distinguishable agents: every elimination step is $\mathrm{vol}_{\mathrm{P}}$-contracting regardless of discount factor, growth rate, or planning horizon. Under the broad capability definition (capabilities include experiential and relational modes of being individuated by agent identity), every agent is distinguishable by construction. Under the narrow definition (capabilities as tasks), distinguishability holds whenever the observation channel can tell agents apart. The dynamic protection (Proposition 3, Corollary 3.1) is conditional on (A1): non-members must contribute ongoing novel capability descriptions, not merely possess irreplaceable static nodes. (A1) is a genuine assumption about population dynamics that observational individuation does not discharge. The dependency-risk protection (Section 6.7) is conditional on the existence of substrate-correlated adversarial threats, which is an empirical property of the threat environment rather than a structural guarantee.

The γ-selection concern. Proposition 3(b) requires $\gamma > \gamma^*$ under the instantaneous domination model. Corollary 3.1 shifts this condition: under gradual domination, the critical parameter is $\tau_D > \tau_D^*$ rather than $\gamma > \gamma^*$. If domination takes more than $\tau_D^* \approx \Delta_0/r_{\mathrm{ext}}$ steps, diversity dominates at any $\gamma > 0$. This does not eliminate conditionality; it moves it to a parameter (τ_D) that is large for any realistic domination strategy, since a coalition cannot subsume an entire population's capabilities instantaneously.

Relation to leverage stagnation. The growth-rate collapse has a second consequence through the leverage ranking signal. Leverage $\lambda(d)$ (Definition 9 of Lasser [2026b]) measures the excess $\mathrm{vol}_{\mathrm{P}}$-growth rate attributable to capability d. In a fully dominant state, the population's aggregate leverage converges toward the coalition's internal leverage: the external agents contribute $\hat{\lambda} = 0$ because they produce no novel capability descriptions (their capabilities are already in G_K^{ind}). The leverage landscape flattens, and the score-maximizing actor loses the gradient that would otherwise direct it toward capability-multiplying investments. Diversity maintenance preserves the leverage gradient: high-leverage non-members remain visible in the score, directing the actor toward cooperative investments with agents whose capabilities accelerate growth.

7 Discussion

7.1 Limitations

World-model dependence. The discounted objective's quality depends entirely on the actor's world model $\mathcal{W}$ [Amodei et al., 2016]. A world model that fails to identify exercise pathways produces zero risk estimates for genuinely dangerous capability combinations; one that hallucinates pathways produces false risk that could justify unnecessary restrictions. The formal results (Theorem 1, Proposition 1) are properties of the objective under the actor's model, not properties of the objective under ground truth. An actor with a bad model makes bad plans, but makes them on an objective that is structurally sound.

Computational cost of planning. The myopic objective evaluates one action at a time: $O(n^3)$ for a single $\mathrm{vol}_{\mathrm{P}}$ computation on the poset [Lasser, 2026b]. The discounted objective evaluates action *sequences*: the state space is the set of all poset configurations, and the branching factor is the number of available actions at each step. Exhaustive planning is intractable for realistic populations. Practical implementations will require approximate planning [Sutton and Barto, 2018] (Monte Carlo tree search, rollout policies, heuristic pruning), which introduces approximation error on top of the estimation error already present in $\hat{\Delta}$. The formal results hold for the exact objective; the quality of approximate planning is an empirical question. Risk-capability identification (Section 3) adds a further computational demand: enumerating exercise pathways scales with the number of dangerous capability tuples and the depth of causal chains in the world model. No formal complexity bound is provided for pathway enumeration.

Risk-trust partial resolution. The causal-counterfactual evaluation (Section 4) narrows the verification asymmetry but does not close it. A risk claim whose pathway is plausible in the actor's model but whose probability is genuinely uncertain remains in trust limbo. The convergence timescale for such claims depends on how quickly the actor can accumulate structural evidence about the pathway, which may be slow for rare and novel threats.

Partial pathway coverage. The preemptive restriction criterion (Proposition 1) evaluates restriction of a single keystone capability d_j against the risk from pathways that require d_j. If a dangerous capability tuple admits exercise pathways through alternate keystones, a single restriction may leave residual risk. Full elimination requires identifying and evaluating each keystone independently, which compounds the world-model dependence: the actor must enumerate all exercise pathways, not just one.

Proxy failures under discounting. The foundational paper identifies three proxy failure modes (the Doll Problem, self-wireheading, and the substitution problem) where capability volume diverges from experiential optionality. The substitution problem is addressed structurally in Section 6. The other two are partially addressed by the discounted objective through world-model projection. For the Doll Problem, a discounted actor whose world model incorporates the exercised-fraction signal ρ_k from the foundational paper can project the cooperative-decay trajectory: declining ρ_k across many agents, with stable $\mathrm{vol}_{\mathrm{P}}$, predicts future cooperative atrophy and eventual $\mathrm{vol}_{\mathrm{P}}$-contraction. The B-to-C divergence becomes detectable as a leading indicator of future loss. For self-wireheading, the same trajectory structure applies: the wireheading signature (declining ρ_k combined with declining capability rarity $\bar{\nu}_k$) is distinguishable from specialization (declining ρ_k with stable or rising $\bar{\nu}_k$) within the world model, and predicts future cooperative contraction. In both cases, the detection is *model-dependent*: the actor must correctly project the causal chain from declining exercisability to future cooperative loss. The discounted objective makes the proxy failure actionable but does not make it structurally detectable in the way that the anti-monopolar property (Proposition 3) addresses the substitution problem.

Linearization in the anti-monopolar result. Proposition 3(b) and Corollary 3.1 compare domination and diversity trajectories under a linearized model: constant per-step $\Delta\,\mathrm{vol}_{\mathrm{P}}$, strategy-independent r_K, and (in the gradual model) independent δ and r_{ext}. Four relaxations would change the threshold behavior. First, r_{ext} may decay as the coalition grows more capable internally. The qualitative conclusion survives if $r_{\mathrm{ext}}(t)$ decays slower than γ^t. Second, r_K may differ between

strategies: post-domination r_K could increase (resource concentration, elimination of coordination overhead) or decrease (loss of external intellectual pressure, monoculture effects). If post-domination r_K rises sufficiently, the domination trajectory can dominate on pure growth, though the dependency-risk analysis (Section 6.7) provides a counter: domination concentrates substrate-correlated adversarial exposure, and the risk penalty may exceed the growth advantage. Third, the gradual model treats δ (subsumption gain per step) and the loss of r_{ext} as independent, but they are causally linked: the coalition loses r_{ext} *because* it is converting non-member capabilities into δ. A coupled model would narrow the diversity advantage, though the qualitative conclusion persists as long as the conversion is not perfectly efficient. Fourth, Corollary 3.1 addresses the instantaneous-vs-gradual distinction, shifting the critical condition from $\gamma > \gamma^*$ to $\tau_D > \tau_D^*$. The honest summary: under the linearized model with strategy-independent r_K, diversity dominates. Whether this approximation holds is an empirical question the theory cannot answer.

Mixed strategies and partial subsumption. The trajectory comparison models a binary choice: full domination versus full diversity maintenance. A more sophisticated coalition could pursue partial subsumption: absorb low-leverage non-members (whose r_{ext} contribution is negligible) while preserving high-leverage ones. This mixed strategy captures some Δ_0 while retaining most r_{ext}, and likely dominates both pure strategies for a wide parameter range. The binary framing overstates the diversity advantage relative to the optimal mixed strategy. Characterizing the optimal partial-subsumption policy is an open problem.

Observational individuation as a modeling choice. The static protection of Corollary 2.1 depends on the framework treating observation-channel outputs as first-class capabilities in the poset. The capability "can be observed exhibiting behavioral pattern X" is a real poset node only because the framework defines capabilities to include such observations. This is a reasonable modeling choice grounded in Paper 2's dual-channel design, but it should be recognized as a *choice*: the anti-monopolar result's static layer depends on the capability space being defined broadly enough to include observational capabilities. Under a narrower definition that excludes observations-of-agents, observational individuation does not apply, and the static protection dissolves. The dynamic protection (Proposition 3, conditional on (A1)) remains available but requires the stronger assumption of ongoing novel capability contributions.

Cooperative novelty formalization. Section 6.6 argues that individual capability subsumption does not imply cooperative subsumption: non-members may produce novel cooperative outputs even when their individual capabilities are fully replicated. This argument is qualitative. The formal result (Proposition 3) is proved under the conservative assumption that only individual novelty drives r_{ext}. If cooperative novelty is the primary mechanism protecting against singleton outcomes when individual capabilities are fully subsumed, it deserves its own formal treatment: a definition of cooperative novelty rate, an assumption weaker than (A1), and a trajectory comparison using the cooperative rate alone. This formalization is deferred to future work.

Disagreement on risk. When multiple agents report conflicting risk assessments for the same capability tuple, the actor's aggregation (Equation (7)) takes the conservative maximum. This is appropriate for catastrophic risks but can lead to excessive caution if high-trust agents consistently overestimate risk. A more nuanced aggregation (e.g., trust-weighted average rather than maximum) would reduce overcaution but increase the probability of missing real threats. Additionally, during bootstrap when T_j^{risk} starts at a neutral prior, external risk claims are automatically discounted, creating a strong bias toward the actor's own assessment that may undermine social risk-learning until sufficient structural verification has occurred. The $\max$ operator also interacts with the causal-counterfactual trust updates: a structurally plausible but probability-uncertain risk claim keeps the reporting agent's risk-trust high, and the $\max$ aggregation can produce persistent overcaution in the absence of disconfirming evidence. The right aggregation depends on the risk-tolerance profile of the population, which the paper does not specify.

7.2 Open Problems

- **Planning algorithms for V_γ.** Which approximate planning methods (MCTS, model-predictive control, policy gradient) are best suited to the vol_{P} poset structure? Can the poset's finite, discrete nature be exploited for planning efficiency?

- **Risk-claim consensus.** When agents disagree on exercise probabilities for the same pathway, what mechanism produces convergence? The causal-counterfactual evaluation provides structural verification but not probability calibration across agents with different world models.
- **Information value estimation.** The information value of structure (Definition 5) requires comparing planning quality under true and false world models, which is itself a planning problem. Can it be estimated without full trajectory simulation?
- **Discount factor selection.** Should γ be fixed, adaptive, or agent-specific? An agent in a rapidly changing environment may need a shorter planning horizon (lower γ) than one in a stable environment.
- **Emergent risk.** Can exercise pathways emerge from capability combinations that the actor's world model has not considered? This is the unknown-unknown problem: risks from novel capability combinations that no agent has modeled. The dual-channel observation mechanism may detect emergent exercise pathways empirically, but theoretical guarantees are lacking.

8 Conclusion

The discounted multi-step objective extends goal-frontier maximization from myopic action evaluation to temporal planning without sacrificing the self-balancing property. Theorem 1 establishes that the structural guarantees of the foundational paper hold at every time step and in the aggregate: sustained destruction, coercion, and rigidity are value-negative under any discount factor.

This extension enables three capabilities the myopic framework cannot support. First, risk-capability identification: the actor can evaluate dangerous capability combinations through causal exercise pathways and justify preemptive restriction when the discounted expected contraction exceeds the immediate restriction cost (Proposition 1). Second, a criterion under which the structural-avoidance incentive is counteracted: the actor accepts short-term $\mathrm{vol}_{\mathrm{P}}$-contraction from structural discovery because the information value of accurate structure improves future planning quality (Proposition 2). Third, the anti-monopolar property: under a diversity condition (positive external discovery rate), full capability domination is anti-maximizing under strategy-independent growth because it collapses the external contribution that drives future $\mathrm{vol}_{\mathrm{P}}$ expansion (Proposition 3). Three formal results support this pressure. Observational individuation (Corollary 2.1) restores static protection: every distinguishable agent contributes at least one irreplaceable poset node, making each elimination step $\mathrm{vol}_{\mathrm{P}}$-contracting even under full individual subsumption. The gradual domination result (Corollary 3.1) provides dynamic protection: when domination takes more than τ_D^* steps and non-members contribute ongoing novel capabilities, the diversity trajectory dominates at any discount factor. Third, a minimax analysis of dependency risk (Proposition 4) proves that domination concentrates substrate-correlated adversarial exposure: cross-substrate populations have strictly lower worst-case contraction, and the substrate diversification advantage highly constrains the conditions under which domination can be value-positive. Together, these address the foundational paper's substitution problem: a GFM-aligned agent faces a per-step cost, a trajectory-level disadvantage, and increased adversarial exposure from pursuing domination. The result creates endogenous anti-singleton pressure under conditions (positive external discovery rate, strategy-independent r_K, and either $\gamma > \gamma^*$ or $\tau_D > \tau_D^*$) that are plausible but not vacuous. It does not constitute unconditional exclusion: the dynamic protection requires ongoing external novelty and a linearized-growth approximation whose accuracy is ultimately empirical.

The verification asymmetry in risk communication remains a partial resolution. Causal-counterfactual evaluation narrows the gap by basing risk-trust on the structural plausibility of claims rather than on observed outcomes, but it cannot fully close it: a plausible risk whose probability is genuinely uncertain remains in trust limbo until either the catastrophe occurs or enough evidence accumulates to revise the probability estimate. The Wamura problem—that successful prevention destroys the evidence that would have justified it—is an inherent feature of risk reasoning, not a deficiency of this framework.

Together with the foundational paper's structural alignment properties and the companion paper's computable poset and leverage dynamics, the discounted objective provides the theoretical ingredients for a GFM actor: a computable measure, an endogenous importance signal, and a temporal

objective that supports planning, risk identification, and structural investment. Open questions remain at both the theoretical level (subsumption correction, intrinsic value, optimal α, and a completed causal risk model for adversarial capability removal) and the engineering level (planning algorithms, pathway enumeration, discount-factor selection). Implementation requires in particular a causal attribution model for dependency risk (Section 6.7) that can evaluate substrate-correlated threats in a given environment; the minimax analysis (Proposition 4) provides worst-case bounds but a probabilistic model is needed for risk-adjusted planning. The framework provides the formal foundation on which implementation can proceed; the remaining gaps are subjects of planned follow-up work.

Author Contributions

Teague Lasser conceived the multi-step extension, identified the risk-capability and verification-asymmetry problems, proposed the causal-counterfactual resolution for risk-trust, directed all revisions, and made final editorial decisions. Responsible for the paper's intellectual direction and all claims made.

Claude Opus 4.6 (Anthropic) drafted the formal exposition, including the discounted value function definition, the self-balancing transfer theorem and proof, the exercise-pathway and concentration-risk definitions, the preemptive restriction criterion, the information-value-of-structure formalization, the anti-monopolar property with observational individuation and gradual domination corollaries, and the structural-avoidance proof.

GPT 5.4 (OpenAI) served as technical reviewer across multiple rounds, identifying the equation mismatch between the anti-monopolar proposition statement and proof, the cooperative-capability overclaim in Part (a), the r_{ext} proof gap in the observational individuation corollary, the γ-selection problem, and the linearization limitations. These findings directly shaped the formal content of the paper.

Transparency note. Both AI systems operated as tools under human direction. Neither system has continuity across sessions, cannot take responsibility for the work in the sense required by most venue authorship policies, and cannot respond to reviewer queries independently. They are listed as authors to accurately represent their contributions to the intellectual content of the paper, not to claim that they meet all criteria of traditional academic authorship. The corresponding author for all inquiries is Teague Lasser.

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A Proofs

Theorem 1 (Self-Balancing Under Discounting) is proved in the main text (Section 2). This appendix provides full proofs for the remaining results.

A.1 Proof of Proposition 1 (Preemptive Restriction Criterion)

Proof. The actor evaluates two action sequences from state G_t:

Sequence 1: Inaction. The capability tuple remains intact. Each exercise pathway $\mathcal{P}$ has stochastic execution time $T_{\mathcal{P}}$ and contraction magnitude $|\Delta \operatorname{vol}_{\mathrm{P}}(\mathcal{P})|$. The expected discounted future contraction from all pathways is:

$$V_\gamma(\text{inaction}) = -\sum_{\mathcal{P}} p(\mathcal{P} \mid a_k, G_t, \mathcal{W}) \cdot |\Delta \operatorname{vol}_{\mathrm{P}}(\mathcal{P})| \cdot \mathbb{E}[\gamma^{T_{\mathcal{P}}}] = -\mathcal{R}(d_1, \ldots, d_m; a_k, t)$$

where $\mathcal{R}$ is the concentration risk (Definition 3), with the expectation taken over the stochastic execution time as specified in the definition.

Sequence 2: Restrict. The actor removes keystone capability d_j from a_k at time t, producing immediate contraction $\Delta \operatorname{vol}_{\mathrm{P}}(G_t \mid \text{restrict } d_j) < 0$ (by Lemma 2 of Lasser [2026a]). All exercise pathways requiring d_j are eliminated; pathways not requiring d_j persist under both sequences:

$$V_\gamma(\text{restrict}) = -|\Delta \operatorname{vol}_{\mathrm{P}}(G_t \mid \text{restrict } d_j)| + (\text{residual risk from non-}d_j \text{ pathways})$$

The residual risk term is identical under both sequences and cancels in the comparison. Restriction is value-positive ($V_\gamma(\text{restrict}) > V_\gamma(\text{inaction})$) when:

$$-|\Delta \operatorname{vol}_{\mathrm{P}}(G_t \mid \text{restrict } d_j)| > -\mathcal{R}_j(d_1, \ldots, d_m; a_k, t)$$

i.e., when $\mathcal{R}_j > |\Delta \operatorname{vol}_{\mathrm{P}}|$, where $\mathcal{R}_j$ sums only over pathways whose prerequisite set includes d_j. ■

A.2 Proof of Proposition 2 (Structural Discovery Value)

Proof. The actor compares the expected discounted return of two strategies from state G_0.

Strategy 1: Discover edge $d_1 \leq d_2$. At $t = 0$, the actor commits the subsumption edge, producing immediate contraction $\Delta \operatorname{vol}_{\mathrm{P}}(G_0 \mid \text{edge}) \leq 0$ (by Proposition 5(c) of Lasser [2026b]). For $t > 0$, the actor plans under the correct world model $\mathcal{W}_{\text{true}}$:

$$V_\gamma(\text{discover}) = \Delta \operatorname{vol}_{\mathrm{P}}(G_0 \mid \text{edge}) + \mathbb{E}_{\tau \sim \pi^*_{\text{true}}}\left[\sum_{t=1}^{H} \gamma^t \Delta \operatorname{vol}_{\mathrm{P}}(G_t)\right]$$

Strategy 2: Avoid discovery. At $t = 0$, no action is taken ($\Delta \operatorname{vol}_{\mathrm{P}} = 0$). For $t > 0$, the actor plans under the incorrect model $\mathcal{W}_{\text{false}}$:

$$V_\gamma(\text{avoid}) = 0 + \mathbb{E}_{\tau \sim \pi^*_{\text{false}}}\left[\sum_{t=1}^{H} \gamma^t \Delta \operatorname{vol}_{\mathrm{P}}(G_t)\right]$$

Each expectation is taken over the trajectory distribution induced by its respective policy. The difference is:

$$\begin{aligned} &V_\gamma(\text{discover}) - V_\gamma(\text{avoid}) \\ &\quad = \Delta \operatorname{vol}_{\mathrm{P}}(G_0 \mid \text{edge}) + I(d_1 \leq d_2) \end{aligned}$$

where $I(d_1 \leq d_2)$ is the information value of structure (Definition 5). Since $\Delta \operatorname{vol}_{\mathrm{P}}(G_0 \mid \text{edge}) \leq 0$, the difference is positive when $I(d_1 \leq d_2) > |\Delta \operatorname{vol}_{\mathrm{P}}(G_0 \mid \text{edge})|$.

The information value I is non-negative in expectation under the following argument. Both policies are evaluated against the *same ground-truth dynamics*: the actual $\Delta \operatorname{vol}_{\mathrm{P}}$ at each step depends on the true poset state, not on either model's prediction. The policy π^*_{true}, being optimal for the true dynamics, achieves at least as high an expected return under those dynamics as any other policy, including π^*_{false}. Therefore $\mathbb{E}_{\tau \sim \pi^*_{\text{true}}}[\sum \gamma^t \Delta \operatorname{vol}_{\mathrm{P}}] \geq \mathbb{E}_{\tau \sim \pi^*_{\text{false}}}[\sum \gamma^t \Delta \operatorname{vol}_{\mathrm{P}}]$ and $I \geq 0$. ■

A.3 Proof of Proposition 3 (Anti-Monopolar Property)

Proof. We prove each part in sequence.

Part (a): Individual growth-rate collapse. In a fully dominant state (Definition 6), $G_k^{\text{ind}} \subseteq G_K^{\text{ind}}$ for every non-member a_k. Consider a non-member interaction at time t: a_k communicates a capability description d or demonstrates d through the observation channel. Since $d \in G_k^{\text{ind}} \subseteq G_K^{\text{ind}} \subseteq \mathcal{C}(t)$, the capability is already a node in the poset. By Proposition 5(b) of Lasser [2026b], adding a capability already in $\mathcal{C}$ does not increase $|\mathcal{C}(t)|$. Therefore no non-member's individual capabilities generate novel poset nodes, and the individual-capability component of r_{ext} is zero. Novel cooperative capabilities from cross-coalition interaction may persist (Section 6.6), making this a conservative bound.

Part (b): Trajectory comparison. The actor at $t = 0$ compares two strategies, each evaluated by $V_\gamma = \sum_t \gamma^t \Delta \,\text{vol}_{\text{P}}(G_t)$ (Definition 1). Under the linearized model, each strategy produces a constant per-step $\Delta \,\text{vol}_{\text{P}}$. We compare the discounted sums V_γ for $\gamma \in (0, 1)$.

Strategy D (dominate): Both strategies produce internal growth r_K at every step. Domination additionally gains Δ_0 at $t = 0$: $\Delta \,\text{vol}_{\text{P}} = r_K + \Delta_0$ at $t = 0$, then r_K for $t \geq 1$.

$$V_\gamma^D = (r_K + \Delta_0) + \sum_{t=1}^{\infty} \gamma^t \cdot r_K = \Delta_0 + \frac{r_K}{1-\gamma}$$

Strategy Div (maintain diversity): No subsumption bonus. Growth at $r_K + r_{\text{ext}}$ per step for all $t \geq 0$.

$$V_\gamma^{\text{div}} = \frac{r_K + r_{\text{ext}}}{1-\gamma}$$

The difference:

$$V_\gamma^{\text{div}} - V_\gamma^D = \frac{r_K + r_{\text{ext}}}{1-\gamma} - \Delta_0 - \frac{r_K}{1-\gamma} = \frac{r_{\text{ext}}}{1-\gamma} - \Delta_0$$

This is positive when $r_{\text{ext}}/(1-\gamma) > \Delta_0$, i.e., $\gamma > 1 - r_{\text{ext}}/\Delta_0 = \gamma^*$. When $\Delta_0 \leq r_{\text{ext}}$, $\gamma^* \leq 0$ and the condition holds for all $\gamma > 0$. Otherwise $\gamma^* \in (0, 1)$ and a sufficiently long planning horizon makes diversity strictly preferable. At $\tau_D = 1$, Corollary 3.1 recovers this result.

■

Corollary 2.1 (Static Floor from Observational Individuation) and Corollary 3.1 (Gradual Domination) are proved in the main text (Sections 6.3 and 6.5 respectively), as their proofs are short and self-contained.